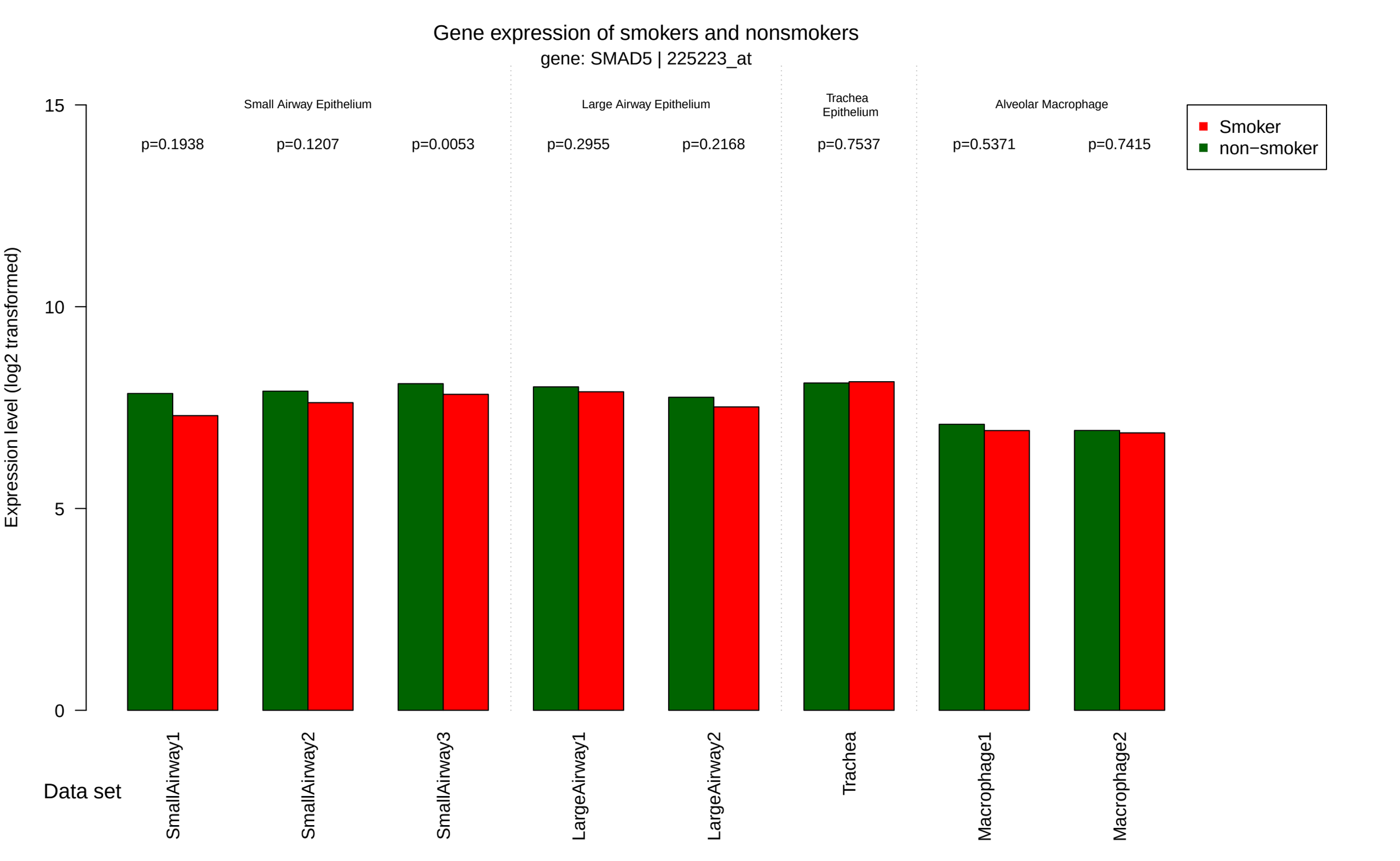
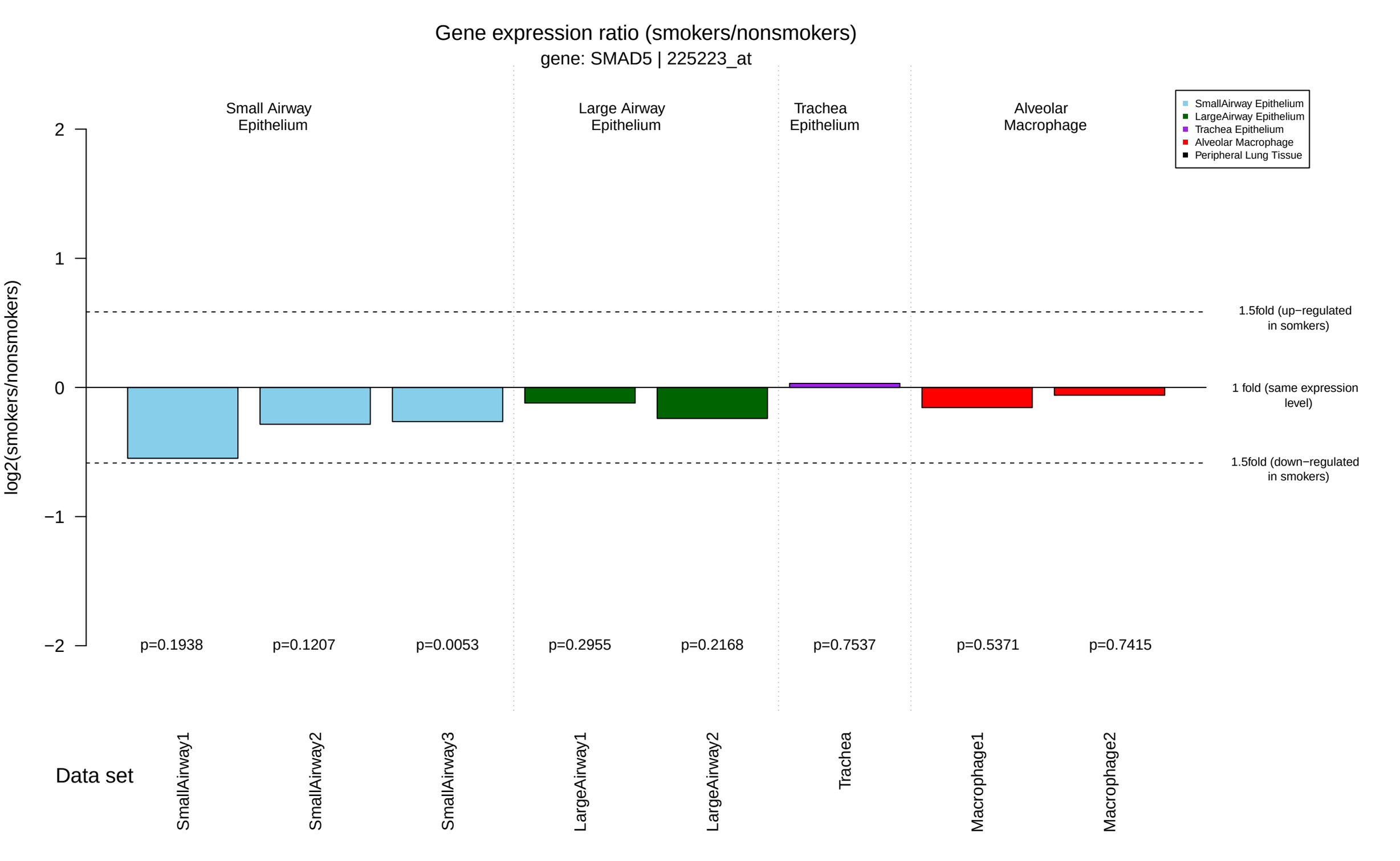
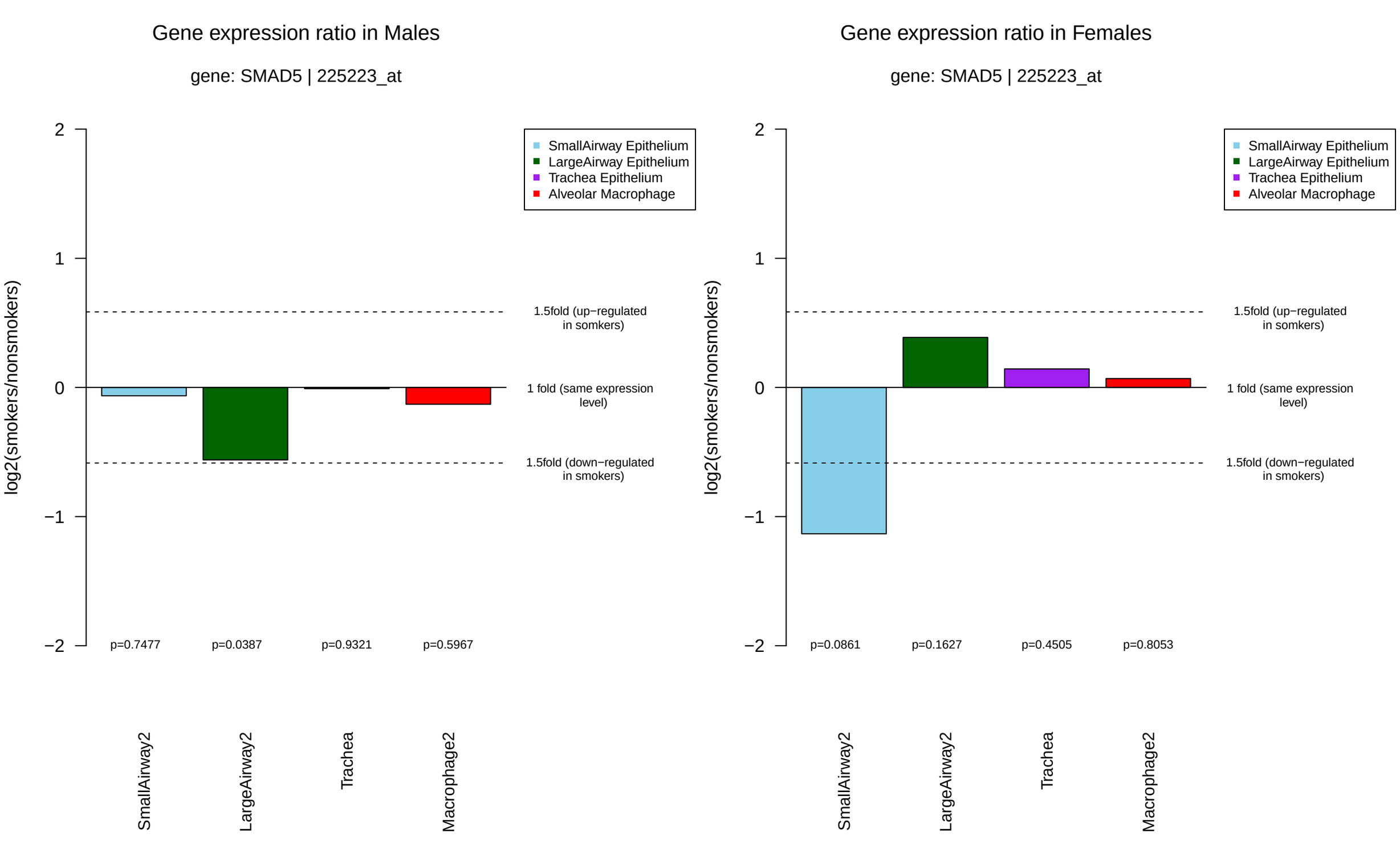
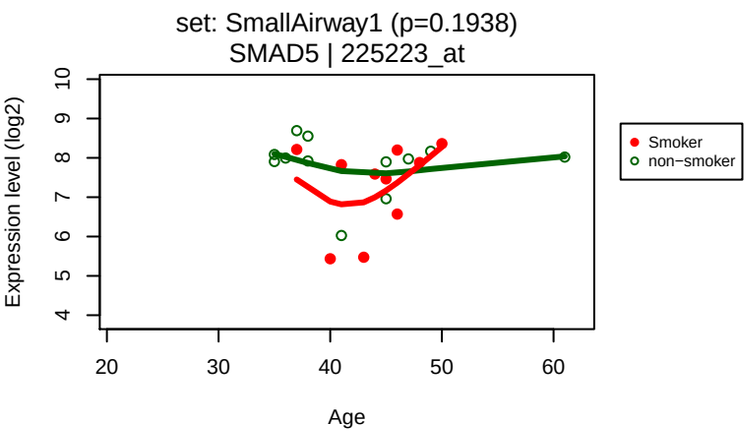
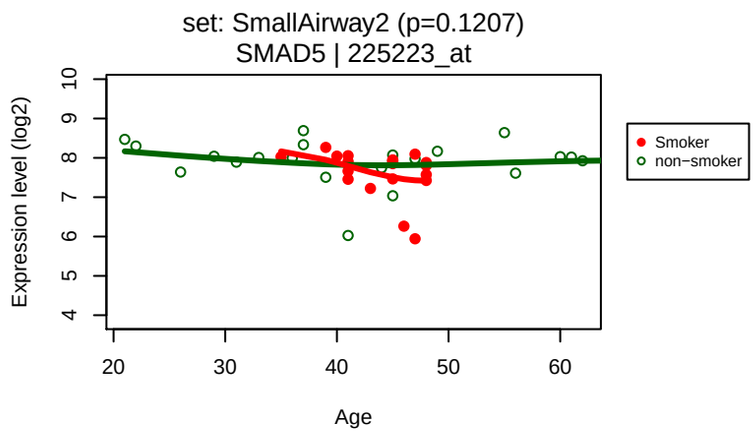
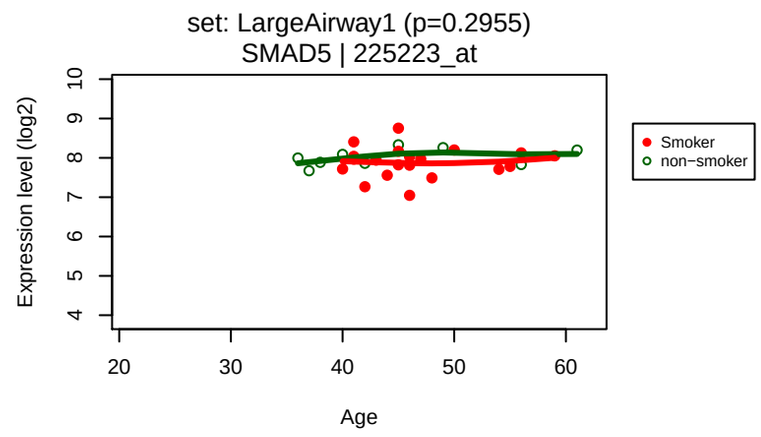

Gene expression of smokers and nonsmokers
gene: SMAD5 | 225223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1938
p=0.1207
p=0.0053
p=0.2955
p=0.2168
p=0.7537
p=0.5371
p=0.7415
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

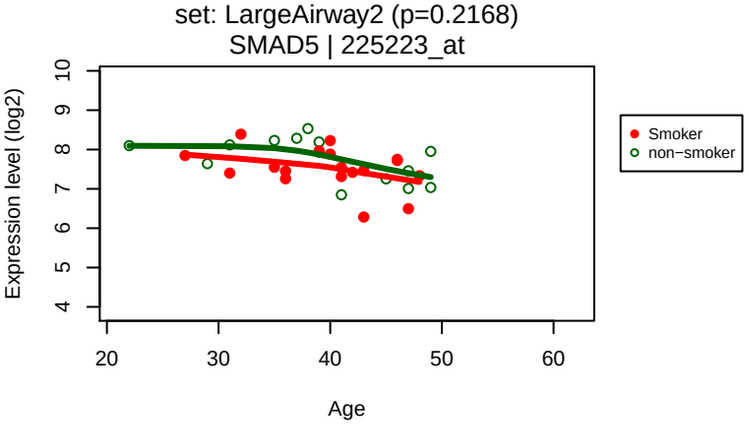

Gene expression ratio (smokers/nonsmokers)
gene: SMAD5 | 225223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1938
p=0.1207
p=0.0053
p=0.2955
p=0.2168
p=0.7537
p=0.5371
p=0.7415
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

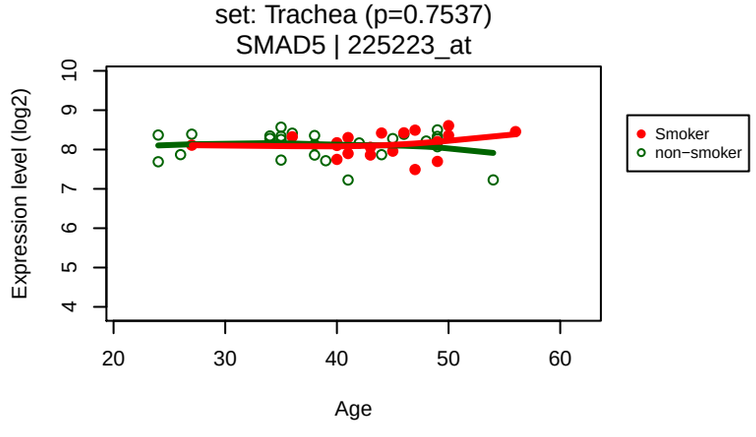

Gene expression ratio in Males
gene: SMAD5 | 225223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7477
p=0.0387
p=0.9321
p=0.5967
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD5 | 225223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0861
p=0.1627
p=0.4505
p=0.8053
SmallAirway2
LargeAirway2
Trachea
Macrophage2

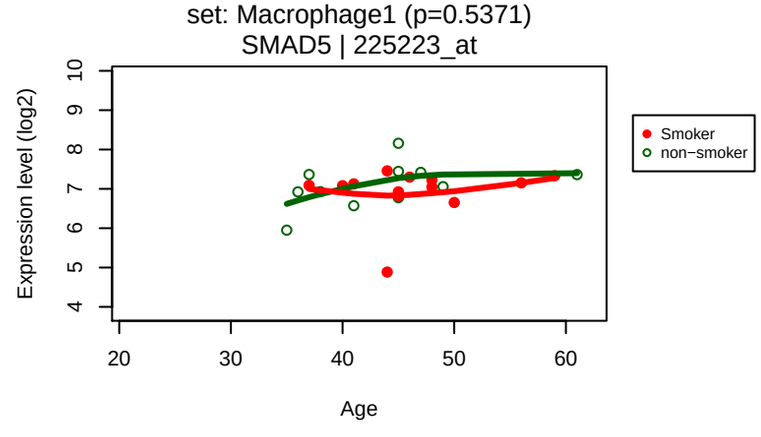

set: SmallAirway1 (p=0.1938)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker

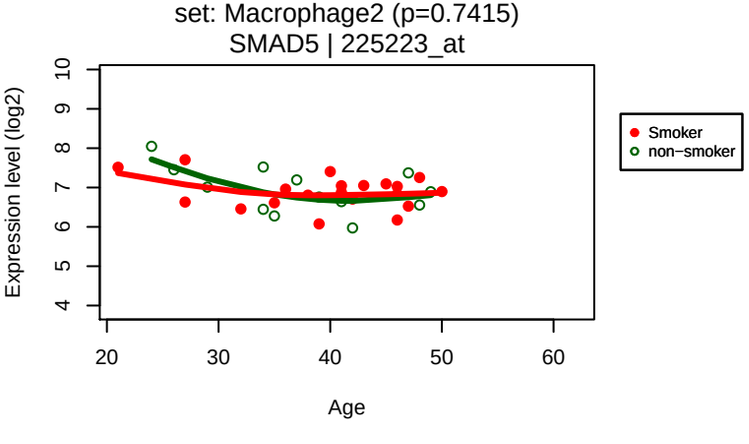

set: SmallAirway2 (p=0.1207)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker

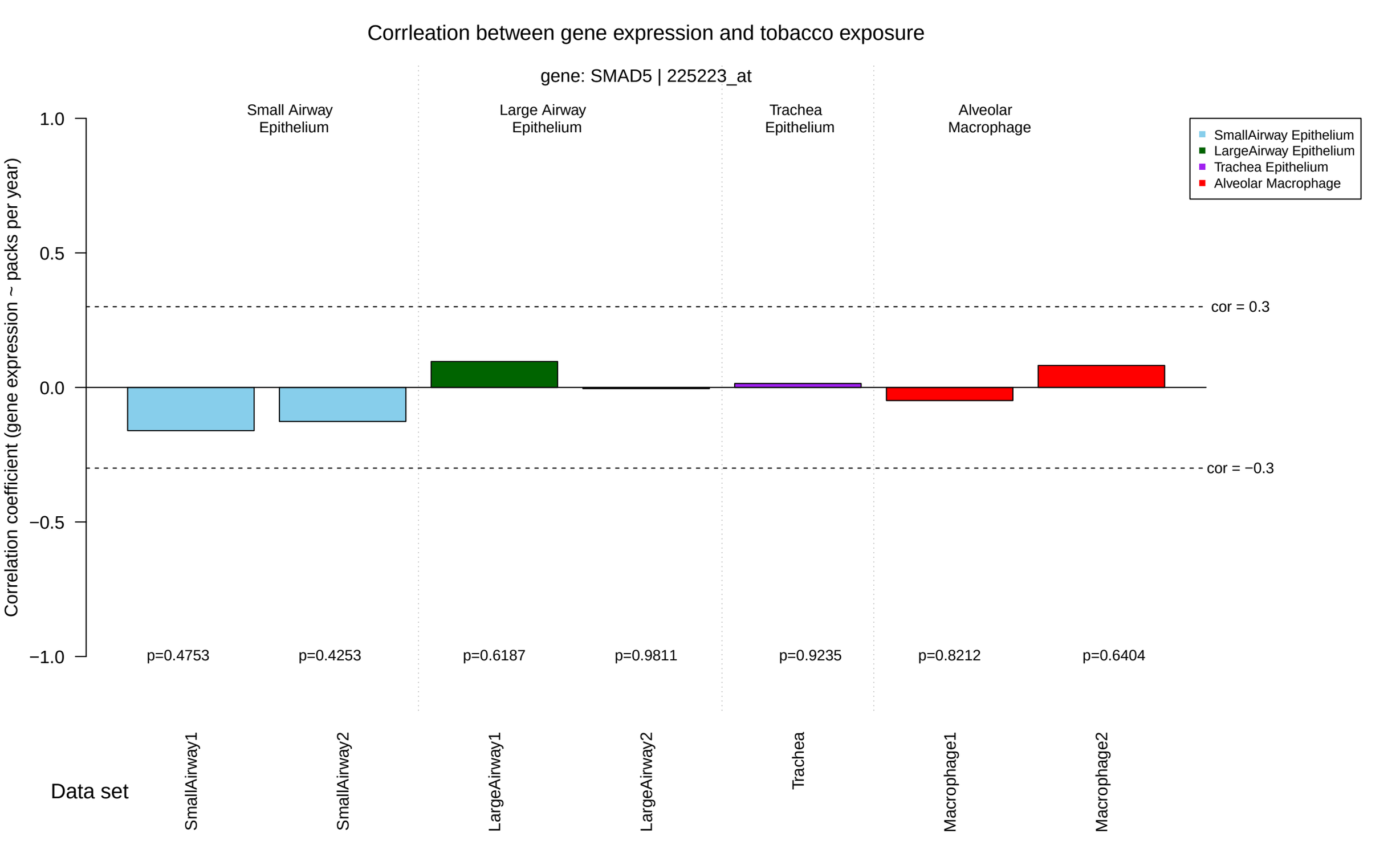

set: LargeAirway1 (p=0.2955)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2168)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7537)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5371)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7415)
SMAD5 | 225223_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD5 | 225223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4753
p=0.4253
p=0.6187
p=0.9811
p=0.9235
p=0.8212
p=0.6404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2